MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
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III SEMESTER B.TECH. (BIOTECHNOLOGY)
END-SEMESTER REGULAR EXAMINATION,22/11/2024 (9:30AM-12:30PM)
SUBJECT: FLUID FLOW OPERATIONS (BIO 2124)
REVISED CREDIT SYSTEM
ANSWER ALL QUESTIONS

TIME: 3 HOURS **MAX. MARKS: 50**

Q. NO	QUESTIONS	M	CO	PO	BTL
1A	A multi-fluid manometer is used to measure the pressure difference between two points, A and B, in a system. The manometer contains three fluids in different segments: Water (ρ_{H2O}=1,000 kg/m^3) at height 0.1 m; Oil (ρ_{Oil}=850 kg/m^3) at height 0.2 m; and Mercury (ρ_{Hg}=13,600 kg/m^3) at height 0.05 m. Point A is connected to the water side of the manometer, and point B to the mercury side. The fluid levels are in equilibrium. i. Calculate the pressure difference $P_A - P_B$ between points A and B. ii. If a small additional pressure is applied at point A, how much would the mercury column height change in the mercury section?	**4**	**1**	**1-3**	**4**
1B	An inclined manometer, set at an angle of 45° with the horizontal, measures the pressure difference between a gas in a container and the atmosphere. The manometer is filled with oil (ρ_{Oil}=850 kg/m^3). The length of the oil column along the inclined tube is 0.15 m, with the oil level higher on the side connected to the container. i. Calculate the pressure of the gas relative to atmospheric pressure. ii. If the inclined angle is increased to 60°, what would be the new pressure reading?	**4**	**1**	**1-3**	**3**
1C	An inclined manometer is used to measure the pressure difference between two points. The angle of inclination can be adjusted. How does the angle of inclination affect the sensitivity of the manometer reading for small pressure changes?	**2**	**1**	**1-3**	**4**
2A	Water with viscosity 0.001 Pa.s flows between two large, parallel plates, separated by a distance of 3 mm. The top plate moves at 1 m/s, while the bottom plate is stationary. Calculate the shear stress on the fluid. If the area of the plate is 0.4 m^2, find the force necessary to keep the plate moving at 1 m/s. How much work is done to maintain the plate movement over 10 seconds?	**4**	**1**	**1-3**	**3**
2B	For a pipe with a diameter of 0.25 m, compare the pressure drop required for a venturi meter and an orifice plate to measure the same flow rate of water at 0.05 m^3/s. Assume the venturi meter has a discharge coefficient of 0.98 and a throat diameter	**4**	**2**	**1-3**	**4**

	of 0.125 m, and the orifice plate has a discharge coefficient of 0.6 and an orifice diameter of 0.1 m. Calculate: i. The required pressure drop for each device. ii. Analyze which device is more efficient for minimizing pressure losses and discuss the implications for energy cost in a fluid transport system.				
2C	In a given flow scenario, two points are selected along a streamline: Point A has a high pressure and low velocity, while Point B has a low pressure and high velocity. Using Bernoulli's equation, analyze the flow behavior between these two points. What factors could lead to this transition?	**2**	**2**	**1-3**	**3**
3A	In a piping system, water flows through a 50-meter straight section of a smooth pipe with a diameter of 0.1 meters. The system also includes two 90° elbows, one valve (half open), and a sudden enlargement where the diameter increases to 0.15 meters. If the flow rate is 0.05 m^3/s, calculate: i. The major loss due to pipe friction. ii. The minor losses due to each fitting. iii. Compare the total minor losses to the major loss. Which loss is more significant in this case? Data: $K_{elbow=}0.9$; $K_{valve}=5$; and $K_{enlargement} = 0.111$.	**4**	**3**	**1-3**	**4**
3B	A system consists of two parallel pipes, Pipe 1 and Pipe 2, with different diameters and lengths. Pipe 1 has a diameter of 0.1 m, a length of 100 m, and a higher Darcy's friction factor (0.02) due to roughness. This pipe also includes an elbow with a minor loss coefficient K=0.5. Pipe 2 has a diameter of 0.15 m, a length of 50 m, and is smoother (with a lower friction factor, 0.1). This pipe only includes a valve with a minor loss coefficient K=1. The flow rate in the system is 0.1 m^3/s, and the total head loss across both pipes is assumed to be the same. Calculate the Flow Distribution between the two Pipes.	**4**	**3**	**1-3**	**4**
3C	For a given pressure drop, discuss the sensitivity of flow rate Q to changes in pipe length L and radius r. How would a small error in estimating pipe radius affect the predicted flow rate, and why is accurate measurement of radius more critical than pipe length in flow calculations?	**2**	**3**	**1-3**	**5**
4A	Consider a packed bed column where a fluid is flowing through a bed of spherical particles. The following parameters are provided: Flow rate = 0.05 m^3/s; Column diameter = 0.5 m; Particle diameter = 0.01 m; Bed length = 3 m; Particle porosity = 0.4; and Fluid properties**:** ρ=1000 kg/m^3; μ=0.001 Pa.s. Calculate the pressure drop across the packed bed.	**4**	**4**	**1-3**	**4**
4B	Calculate the shape factor of a cylindrical object whose length is twice the diameter.	**2**	**4**	**1-3**	**4**
4C	To ensure turbulent conditions and minimum mixing time during agitation with a turbine impeller, the Reynolds number must be at least 10^4. i. A stirred laboratory-scale fermenter with a turbine impeller 5 cm in diameter is operated at 800 rpm. If the density of broth being stirred is close to that of	**4**	**4**	**1-3**	**5**

	water, what is the upprer limit for viscosity of the suspension if adequate mixing is to be maintained? ii. The mixing system is scaled up so the tank and impeller are 15 times the diameter of the laboratory equipment. The stirrer in the large vessel is operated so that the stirrer tip speed (tip speed = $\pi N_i D_i$) is the same as in the laboratory apparatus. How does scale-up affect the maximum viscosity allowable for maintenance of turbulent mixing conditions?				
5A	A fluidized bed consists of spherical particles of diameter d_p=0.5 mm, and density ρ=2500 kg/m^3. The bed height at minimum fluidization is h=0.25 m, and the bed voidage is ε=0.45. The fluidizing medium is air at room temperature with a density ρ=1.2 kg/m^3 and viscosity $\mu=1.8\times10^{-5}$ Pa.s. Calculate the superficial velocity at the minimum fluidization condition. **Hint:** Use Ergun's equation for calculating the superficial velocity V_{mf} at minimum fluidization condition.	**4**	**4**	**1-3**	**4**
5B	A tower having a diameter of 0.1524 m is being fluidized with water at 20.2°C. The uniform spherical beads in the tower bed have a diameter of 4.42 mm and a density of 1603 kg/m^3. Estimate the minimum fluidization velocity and compare with the experimental value of 0.02307 m/s of Wilhelm and Kwauk.	**4**	**4**	**1-3**	**4**
5C	How does particle size and density affect the fluidization behaviour in a reactor?	**2**	**4**	**1-3**	**5**
	CO: Course Outcome; BLOOM TAXONOMY LEVEL: 1-Remember, 2-Understanding, 3-Application, 4-Analyzis, 5-Evaluation, 6-Creation				